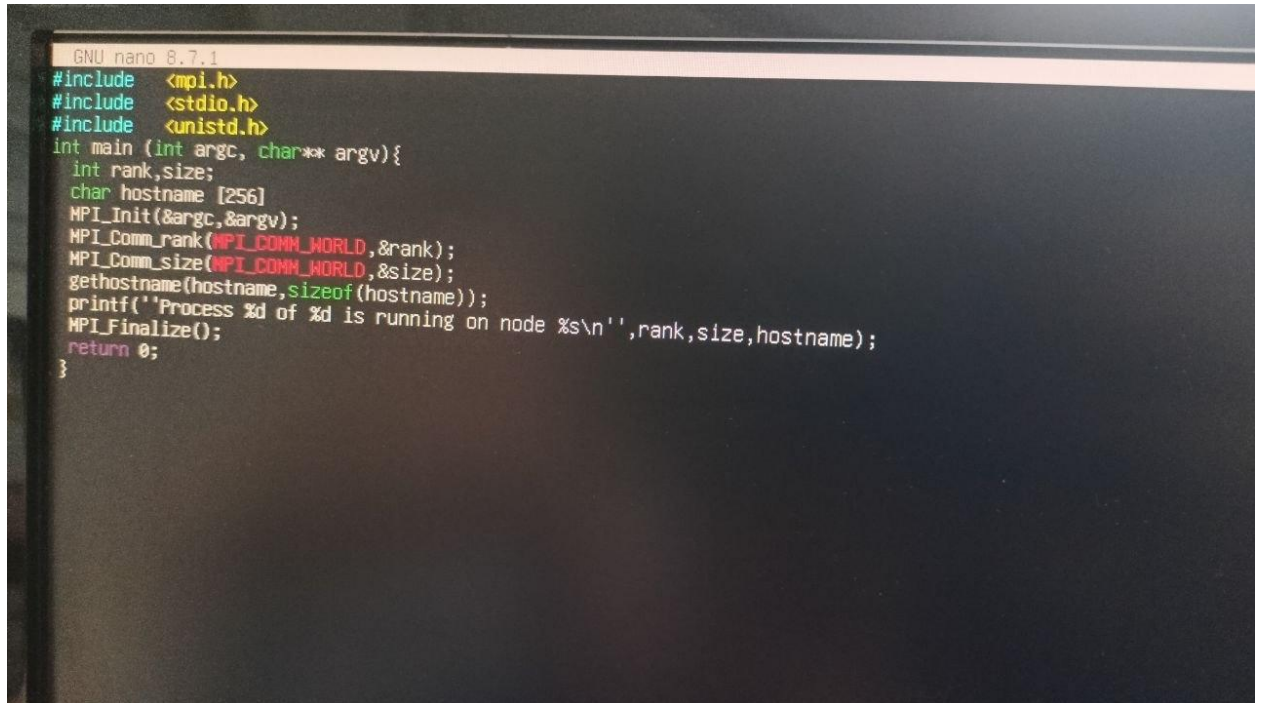


Звіт КІ 24-1, Мельник Є. В.

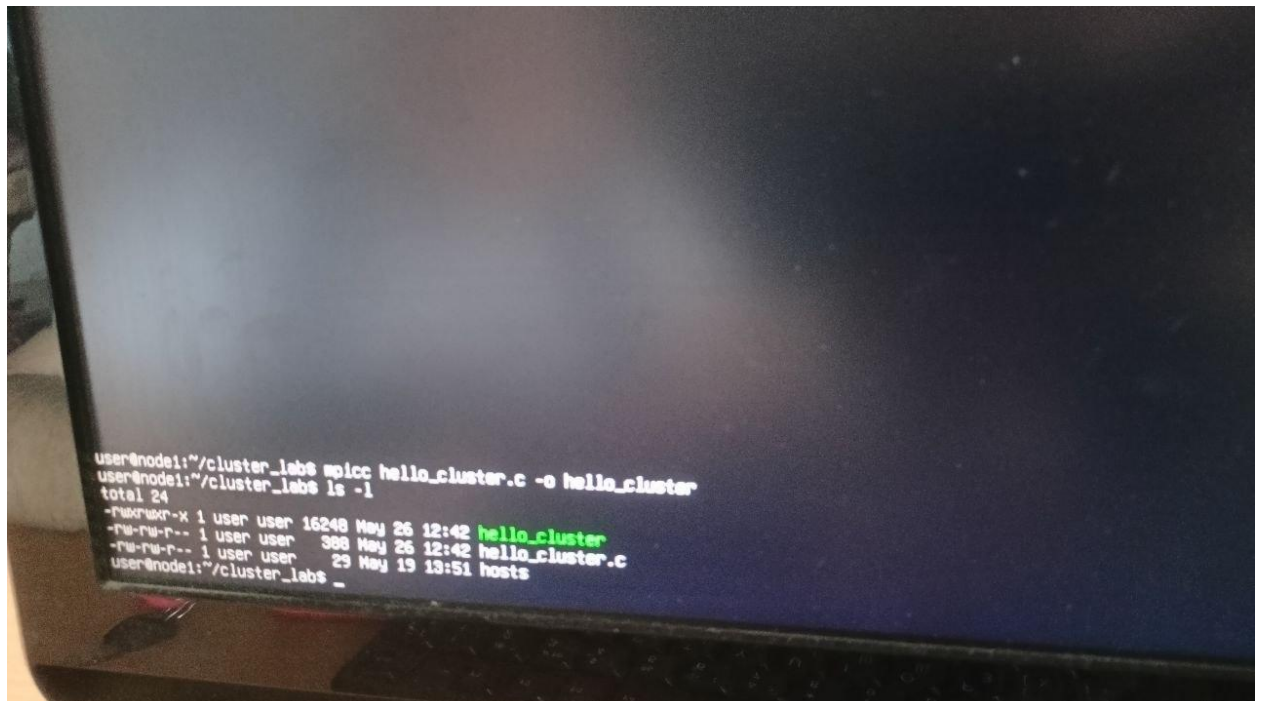
Лабораторна робота №5

Тестова програма



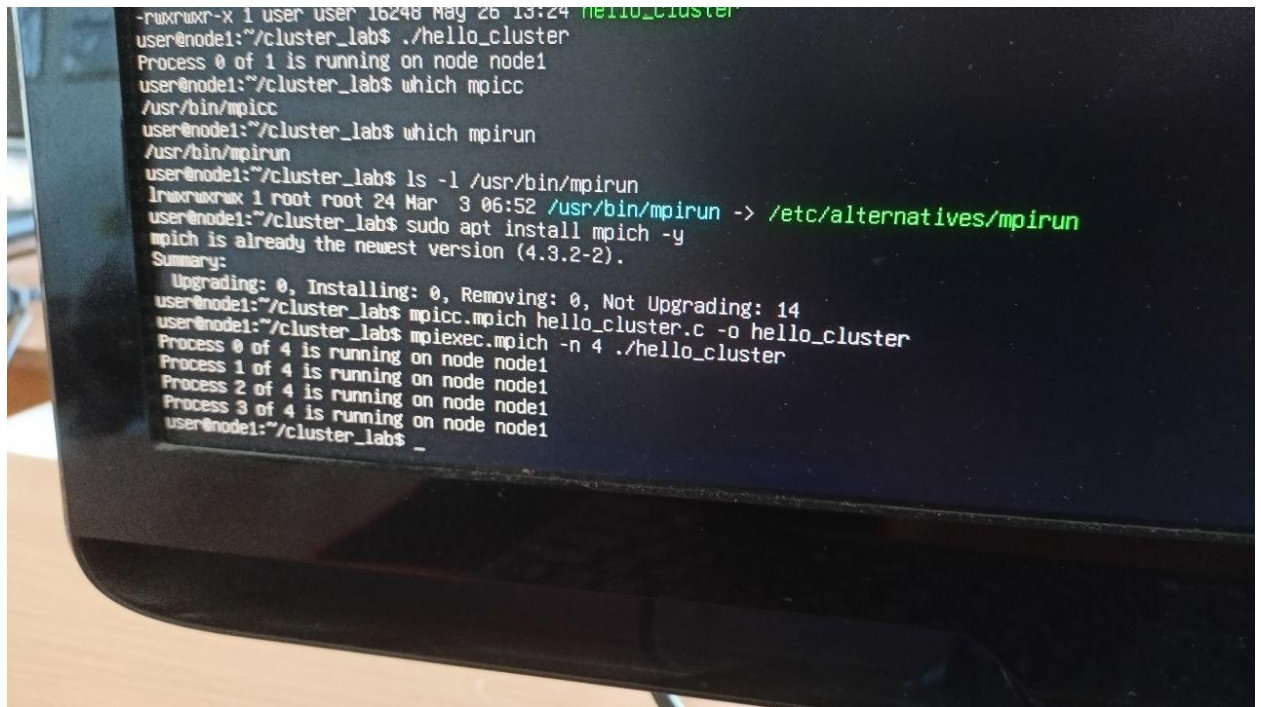
```
GNU nano 8.7.1
#include <mpi.h>
#include <stdio.h>
#include <unistd.h>
int main (int argc, char** argv){
    int rank,size;
    char hostname [256]
    MPI_Init(&argc,&argv);
    MPI_Comm_rank(MPI_COMM_WORLD,&rank);
    MPI_Comm_size(MPI_COMM_WORLD,&size);
    gethostname(hostname,sizeof(hostname));
    printf("Process %d of %d is running on node %s\n",rank,size,hostname);
    MPI_Finalize();
    return 0;
}
```

Компіляція програми



```
user@node1:~/cluster_lab$ mpicc hello_cluster.c -o hello_cluster
user@node1:~/cluster_lab$ ls -l
total 24
-rwxrwxr-x 1 user user 16248 May 26 12:42 hello_cluster
-rw-rw-r-- 1 user user 388 May 26 12:42 hello_cluster.c
-rw-rw-r-- 1 user user 29 May 19 13:51 hosts
user@node1:~/cluster_lab$
```

Запуск програми

A photograph of a computer monitor displaying a terminal window. The terminal shows a series of commands and their outputs. The user is at a prompt 'user@node1:~/cluster_lab\$'. They run './hello_cluster', which outputs 'Process 0 of 1 is running on node node1'. Then they run 'which mpicc', which outputs '/usr/bin/mpicc'. Next, they run 'which mpirun', which outputs '/usr/bin/mpirun'. Then they run 'ls -l /usr/bin/mpirun', which outputs 'lrwxrwxrwx 1 root root 24 Mar 3 06:52 /usr/bin/mpirun -> /etc/alternatives/mpirun'. Then they run 'sudo apt install mpich -y', which outputs 'mpich is already the newest version (4.3.2-2)'. Then they run 'mpicc -x 1 user user 16248 May 26 13:24 hello_cluster', which outputs 'Summary: Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14'. Then they run 'mpicc.mpich hello_cluster.c -o hello_cluster', which outputs 'Process 0 of 4 is running on node node1', 'Process 1 of 4 is running on node node1', 'Process 2 of 4 is running on node node1', and 'Process 3 of 4 is running on node node1'. Finally, they run 'mpirun --hostfile hosts -np 4 ./hello_cluster', which outputs 'user@node1:~/cluster_lab\$ _'.

```
-rwxrwxr-x 1 user user 16248 May 26 13:24 hello_cluster
user@node1:~/cluster_lab$ ./hello_cluster
Process 0 of 1 is running on node node1
user@node1:~/cluster_lab$ which mpicc
/usr/bin/mpicc
user@node1:~/cluster_lab$ which mpirun
/usr/bin/mpirun
user@node1:~/cluster_lab$ ls -l /usr/bin/mpirun
lrwxrwxrwx 1 root root 24 Mar 3 06:52 /usr/bin/mpirun -> /etc/alternatives/mpirun
user@node1:~/cluster_lab$ sudo apt install mpich -y
mpich is already the newest version (4.3.2-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14
user@node1:~/cluster_lab$ mpicc.mpich hello_cluster.c -o hello_cluster
Process 0 of 4 is running on node node1
Process 1 of 4 is running on node node1
Process 2 of 4 is running on node node1
Process 3 of 4 is running on node node1
user@node1:~/cluster_lab$ _
```

Команда ***mpirun --hostfile hosts -np 4 ./hello_cluster*** не працювала через помилку яку я не зміг пофіксувати.

Попередні пункти такі як встановлення другої ос, налаштування ір адрес та ssh сервера були виконані успішно але не задокументовані.